

Chuong Nguyen

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RESEARCH INTERESTS

Large language models; LLM reasoning through human-inspired approaches; agentic frameworks; data mining; LLM-assisted qualitative data analysis; context awareness, agent learning and navigation, and forecasting.

EDUCATION

University of California, San Diego

B.S. in Computer Science, Department of Computer Science & Engineering | GPA: 3.803

Sep 2022 - Jun 2026

PUBLICATIONS

* denotes co-first authors.

Conference Papers

[2026] Xinyu Pi*, Qisen Yang*, **Chuong Nguyen***, Hua Shen. *Position: Bridge Human Interpretation and Machine Representation With Explicit Specification For Qualitative Data Analysis In LLM Era*. International Conference on Machine Learning (ICML), Position Paper Track, 2026. arXiv:2601.11739.

[2026] Taggert Smith*, **Chuong Nguyen***, Qisen Yang, Oishani Bandopadhyay, Yaru Su, Nadia Polikarpova, Xinyu Pi. *QualAlign: Benchmarking Automated Qualitative Coding Against Human Schemas*. Conference on Language Modeling (COLM), 2026. openreview.net/forum?id=FaPyCwiiTo.

[2024] Andrew Smithwick*, **Chuong Nguyen***, Emily Gorial*, Natasha Tran*, A. M. Flores, I. N. S. Munyaka. *"Parent seeking Roblox Safety Help": Comparing Parental Roblox Concerns to Roblox Offerings*. IEEE International Symposium on Technology and Society (ISTAS), pp. 1-9, 2024.

Preprints

[2025] Xinyu Pi*, Qisen Yang*, **Chuong Nguyen***. *LOGOS: LLM-driven End-to-End Grounded Theory Development and Schema Induction for Qualitative Research*. arXiv preprint arXiv:2509.24294, 2025.

CURRENT PROJECTS

- Episteme: deep research system (manuscript in preparation).
- Coreference resolution (in progress).

PRESENTATIONS

- Poster: QualAlign: Benchmarking Automated Qualitative Coding Against Human Schemas. COLM 2026, San Francisco, CA, Oct 2026.
- Poster: Position: Bridge Human Interpretation and Machine Representation With Explicit Specification For Qualitative Data Analysis In LLM Era. ICML 2026, Seoul, South Korea, Jul 2026.
- "Parent seeking Roblox Safety Help". Early Research Scholars Program Symposium, UC San Diego, 2024.
- Credit Card Fraud Detector. ACM AI Symposium, UC San Diego, 2023.

RESEARCH EXPERIENCE

Research Assistant, Halicioglu Data Science Institute (HDSI) - UC San Diego

Apr 2025 - Present

- Research on large language models, agentic frameworks, and data mining for qualitative research, including grounded theory development, schema induction, and human-aligned reasoning.
- Fine-tuned LLMs with SFT and GRPO; explored test-time scaling and process reward models.
- Co-first author on ICML 2026 position paper and COLM 2026 paper (QualAlign); co-first author on LOGOS.

Research Assistant, Ujima Lab - UC San Diego

Sep 2023 - Oct 2024

- Beauty Filter: analyzed facial enhancements from three filter applications (Snatched, Blochi Gora, Baby Girl) to investigate cultural biases in appearance modification; migrated the facial landmark analysis pipeline to a remote server for parallel processing at scale.
- Smart Mirror: implemented DeepFace and Retina for facial recognition and racial classification on a Raspberry Pi smart mirror, delivering real-time analysis for a public art installation on racial bias.
- Roblox Safety: co-first author on "Parent seeking Roblox Safety Help" (IEEE ISTAS 2024).

Early Research Scholars Program (ERSP) - UC San Diego

Sep 2023 - Jun 2024

- Conducted topic modeling on 10,000+ Reddit posts and surveyed 100 participants to identify four primary parental concerns on Roblox; proposed actionable platform-safety recommendations.

- Built a hate-crime news pipeline with SerpAPI, NewsPlease, DBSCAN, MeanShift, and the OpenAI API.

PROJECT EXPERIENCE

Software Developer, Zooseeker

- Designed and implemented the database schema for a zoo navigation Android application.
- Revamped the Android UI and maintained project documentation for the development team.

Apr 2024 - Jun 2024

ACM AI - UC San Diego
Credit Card Fraud Detector

- Compared KNN, Random Forest, Decision Tree, SVM, and Logistic Regression for fraud detection.
- Built and presented a fraud detection web application at the ACM AI symposium.

Oct 2022 - Jan 2023

TEACHING & MENTORING

Peer Mentor, UCSD Mentor Collective

2024

HONORS & AWARDS

- Provost Honors - UC San Diego
- Machine Learning Specialization - Coursera

LEADERSHIP & SERVICE

Volunteer Youth Leader, Vietnamese Eucharistic Youth Movement

- Teach and mentor young children in the youth program.

2022 - Present

SKILLS

Programming: Python, Java, C++, SQL

ML & Data: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, OpenCV, Streamlit

Research: NLP & LLMs, LLM fine-tuning (SFT, GRPO), RAG, reinforcement learning, agentic frameworks, data mining, topic modeling, qualitative methods, computer vision

Tools: Git, Linux, LaTeX

Languages: English, Vietnamese